

Multiple Linear Regression Project Documentation

1. Introduction

Multiple Linear Regression (MLR) is a supervised machine learning technique used to predict a target variable (dependent variable) based on **two or more independent features**. In this project, our goal is to predict an individual's **income** based on their **age** and **experience**.

2. Dataset Overview

This dataset considers multiple features to accurately estimate income:

- **Features (Independent Variables):**
 - **age**: Age of the person (in years).
 - **experience**: Work experience (in years).
- **Target Variable (Dependent Variable):**
 - **income**: Annual or monthly income of the person.

The dataset is relatively small, containing a total of **20 rows** (out of which **16 rows** are used for training and **4 rows** for testing).

3. Step-by-Step Code Walkthrough & Implementation

Step 1: Importing Libraries

Necessary Python libraries are imported for analysis and modeling:

- **numpy** and **pandas**: For data handling and numerical operations.
- **matplotlib.pyplot** and **seaborn**: For data visualization and plotting graphs.
- **sklearn.model_selection (train_test_split)**: To split the dataset into training and testing sets.
- **sklearn.linear_model (LinearRegression)**: To train the MLR model.
- **sklearn.metrics (mean_squared_error, r2_score)**: To evaluate model performance.

Step 2: Loading the Dataset

The CSV dataset is loaded using the Pandas library, and the structure is inspected using the first few rows (**df.head()**):

```
Python
df =
pd.read_csv("/kaggle/input/datasets/farhankarim11/multiple-linear-regression-project/multiple_linear_regression_dataset.csv")
```

Step 3: Splitting Features and Target

Here, **age** and **experience** are set as the independent features (X), while **income** is set as the target variable (y):

```
Python
X = df[['age', 'experience']]
y = df['income']
```

Step 4: Train-Test Split

To check the model's generalization capability, the data is split into 80% training and 20% testing sets (**test_size=0.2**):

- **Train data shape:** (16, 2)
- **Test data shape:** (4, 2)

Step 5: Model Training (Fitting)

The **LinearRegression()** class is initialized and fitted on the training data (**X_train, y_train**):

```
Python
model = LinearRegression()
model.fit(X_train, y_train)
```

4. Model Results & Interpretation

After training the model, we obtain the following parameters:

- **Intercept (\$b_0\$):** 31465.06
- **Coefficients (\$b_1, b_2\$):** [-101.06, 2154.81]

Equation Form:

$$\text{Income} = 31465.06 - (101.06 \times \text{age}) + (2154.81 \times \text{experience})$$

- **Effect of Experience:** The experience coefficient indicates that for every additional year of experience, income increases by approximately **\$2,154.81** (holding other factors constant).
- **Effect of Age:** The age coefficient reflects the specific pattern captured within this small dataset.

5. Model Evaluation Metrics

After making predictions on the test data (**X_test, y_test**), the model's accuracy was evaluated:

- **Mean Squared Error (MSE):** 753796.77 — Represents the average squared difference between the estimated values and the actual value.
- **R-squared Score (\$R^2\$):** 0.94 (i.e., **94%**) — An excellent score indicating that our model explains 94% of the variance in the data.

6. Data Visualization

A scatter plot was created between **Actual Income** and **Predicted Income** to visually inspect model performance. The red dashed line ($y = x$) represents perfect predictions, showing how closely aligned our predictions are with the actual values.

7. Conclusion

This Multiple Linear Regression model performs exceptionally well with an R^2 score of **0.94**. This proves that age and experience can reliably predict income levels within this context. Future improvements could involve using larger datasets and advanced algorithms (such as Random Forest or Gradient Boosting).